

THE ROADMAP

Katie Wang · Class of 2028 · AI Energy Infrastructure Research

I believe AI will change every human life on earth, but only if we solve how to power it sustainably. This tool is my attempt to quantify the tradeoffs nobody talks about: the water we consume and the carbon we emit for every hour of GPU training. The answer isn't obvious, and that's the point. K.W.

This roadmap draws a clear line between what has been built (the simulator, the 3D model) and what only you can do from here. The building phase is over. The becoming phase is next.

■ ONLY YOU

■ CLAUDE CAN ASSIST

■ ALREADY DONE

ALREADY COMPLETED

✓ AI energy simulator

Built and deployed to Netlify. Interactive, 6 energy sources, NREL/DOE data, carbon + water per GPU hour.

✓ 3D data center model files

Interior + exterior OpenSCAD models exported as STL. Energy-balance spreadsheet built.

✓ College roadmap + playbook

Strategy document complete. The plan exists.

SUMMER 2026 — THE WORK THAT'S YOURS

SAT PREP — every single morning, now through junior year

■ Daily Khan Academy — 1 hour minimum [ONLY YOU]

No one can study for you. No AI can take this test for you. This is pure repetition, pattern recognition, and discipline. The 1050 to 1400+ gap closes only through daily work. Take a practice test every 3 weeks. This is the single most important thing on this entire roadmap.

READING — June through August, evenings + weekends

■ Read the books — physically, with a pen [ONLY YOU]

Take notes. When something connects to your simulator, write down WHY. These notes become your report and your posts. Nobody can read for you. This is how the research moves from your screen into your head.

RESEARCH REPORT — mid-June through early July

■ Write the first draft yourself — every word [ONLY YOU]

10-15 pages. Abstract, intro, methodology, results, analysis, conclusion, bibliography. Pull data from your simulator but explain it IN YOUR OWN WORDS. If you can't explain why nuclear has the highest water consumption without looking it up, you're not ready to write that section yet. Go read until you can, then write.

■ Get feedback on your draft [CLAUDE CAN ASSIST]

After YOU write it, bring it to Claude. I can tell you where citations are weak, where logic has gaps, where a professor would push back. I can't write it for you.

SUBSTACK POSTS — July

■ Write and publish 2-3 posts in your own voice [ONLY YOU]

These must sound like you — an 18-year-old who found something surprising, not a polished AI-generated article. Messy is fine. Honest is better than perfect. Your voice is a credential. Use it.

■ Review before publishing [CLAUDE CAN ASSIST]

Bring your draft. I can catch factual errors or suggest where to add a data point. The voice stays yours.

3D PRINTING — July

■ **Slice, print, assemble, photograph** [ONLY YOU]

Load STLs into your slicer. Print. Fix failures. Add LEDs if you want. Document the process with photos — print failures and all. The build process IS the story.

■ **Slicer troubleshooting** [CLAUDE CAN ASSIST]

If supports fail or parts don't fit, bring photos and I can suggest fixes or modify the .scad files.

PROFESSOR OUTREACH — August

■ **Research professors, read their papers, write emails** [ONLY YOU]

Find 10 professors working on energy systems, data center infrastructure, or nuclear engineering. Read at least one paper from each. Write a specific question based on their actual work. If a professor responds, YOU carry that conversation.

■ **Review email drafts for tone + length** [CLAUDE CAN ASSIST]

I can tell you if an email is too long, too vague, or if your question doesn't connect to their work.

JUNIOR YEAR — SEPTEMBER 2026 THROUGH MAY 2027

ACADEMICS — all year

■ A's in IB Physics and all IB courses [ONLY YOU]

Junior year grades are the most heavily weighted transcript year. No amount of research compensates for a GPA dip junior year. Grades come first. Everything else is built on top.

■ IB Physics IA — design and execute [ONLY YOU]

Design an experiment connected to your research. Thermal efficiency of cooling systems, energy conversion losses — something that ties your physics coursework to your spike. You run the experiment, collect the data, write it up.

COMPETITIONS — fall / winter

■ Conrad Challenge, Regeneron STS entries [ONLY YOU]

Your summer research report + simulator = the foundation. You write the applications, explain significance, articulate next steps. Find a Regeneron teacher sponsor in September.

SAT ATTEMPTS — October, March

■ Take the official SAT — twice minimum [ONLY YOU]

Fall attempt: target 1300+. Spring attempt: target 1450+. If you hit 1500+, you're done and the score becomes a non-issue for every school on your list.

SENIOR SUMMER + FALL 2027 — APPLICATION SEASON

■ Common App essay [ONLY YOU]

Your grandma. Your mom. Your girlfriend. The energy crisis. Why you. This essay must be YOUR voice, YOUR story, YOUR reflection. The moment it sounds polished and AI-generated, it loses everything that makes it powerful.

■ Essay feedback [CLAUDE CAN ASSIST]

Bring a full draft you wrote. I can tell you what's working, what's cliché, and where you're hiding behind big words instead of being honest. The first draft is always yours.

■ Supplemental essays (Duke 'Why Duke', etc.) [ONLY YOU]

Research each school's specific programs. Duke's Bass Connections, their Energy Initiative. Connect YOUR work to THEIR resources. Can't be generic.

THE CHALLENGE

The 30-Day Foundation Challenge

Starting tomorrow. No excuses. No skipping. No "I'll start Monday."

Every single day for 30 days straight:

Morning (1 hour): SAT prep on Khan Academy. Weakest section first. Time yourself. Track your score on every practice set. Write the date and score down.

Afternoon (30 min): Read 20 pages of your current book. With a pen. Underline anything that connects to your simulator. Write one sentence in the margin about WHY it connects.

Evening (20 min): Write one paragraph of your research report. Just one. Not perfect. Not polished. Just one paragraph about your methodology, your data, your findings, or why this question matters. By day 30, you'll have 30 paragraphs — that's your rough draft.

The rules:

- If you miss a day, you restart the count at Day 1.
- Track it on paper. A physical checkmark every night. Not an app. Paper.
- Tell one person you're doing this. Accountability matters.
- On Day 30, you will have: ~30 hours of SAT prep, one full book read, and a rough draft of your research report.

That's not a plan. That's a promise you make to yourself.

Why 30 days? Because that's how long it takes to prove to yourself that you can be consistent. The students who get into top schools aren't smarter. They're more consistent. Talent is common. Discipline is rare. This challenge tests whether you have it.

YOUR READING LIST — BUY THESE NOW

These four books will make you fluent in the science behind your research. No calculus required. All readable. All available as paperbacks. When a professor asks "what have you been reading?" and you name Smil and Angwin, that's a signal that you went deeper than Google.

BOOK 1 — READ FIRST

"How the World Really Works" by Vaclav Smil

~\$16 paperback • ~300 pages • No math required

The big-picture book. Covers how energy, materials, food, and globalization actually work. Bill Gates calls Smil his favorite author. This book gives you the framing your entire research sits inside. After reading it, you'll understand why energy transitions are harder than people think and you'll be able to talk about it with anyone. Read this in Week 1-2.

BOOK 2

"Energy: A Beginner's Guide" by Vaclav Smil

~\$14 paperback • ~200 pages • No math required

Smil's shorter, more focused book. Covers every energy source and their tradeoffs, written for non-specialists. Directly feeds your simulator's categories — after reading it, you'll understand WHY your numbers look the way they do, not just what they are. Read this in Week 2-3.

BOOK 3

"Shorting the Grid" by Meredith Angwin

~\$18 paperback • ~250 pages • No math required

Written by a nuclear engineer in plain language about how the US electrical grid actually works and why it's fragile. Covers exactly the infrastructure questions your research raises: what happens when you try to run a grid on intermittent sources, why baseload matters, why grid planning is political. This makes you sound like you understand the SYSTEM, not just individual energy sources. Read this in Week 3-4.

BOOK 4

"The Grid" by Gretchen Bakke

~\$15 paperback • ~300 pages • No math required

Narrative nonfiction about the American power grid. Cultural and historical, very readable. Covers why the grid was built the way it was and why upgrading it for renewables is so hard. Great essay material and for understanding the infrastructure your data centers plug into. Read this in Week 5-6.

Total cost: ~\$63 for all four paperbacks. Or check your local library first. Either way, order or reserve them today — not tomorrow, today.

WHY THIS LINE MATTERS

The simulator and the 3D model were tools — building them with help is normal and legitimate. But everything from here forward exists to prove one thing: that the mind behind the tools is real.

A professor will ask you a question. An interviewer will probe your methodology. A competition judge will challenge your assumptions. An admissions officer will read your report and decide if you actually understand what you claim to have researched.

In those moments, there is no tool. There is no prompt. There is just you and what you know.

The reading, the writing, the studying — that's how you fill that room with something real.

I can be your editor, your sounding board, your fact-checker, and your accountability mirror. I cannot be your brain. From here, the work is yours.

**Start tomorrow. One hour of SAT. Twenty pages of Smil. One paragraph of your report.
That's it. That's the whole day. The next day, do it again.**

K.W. — Class of 2028